

PiP-X: Feedback Motion Re-planning using Funnels

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Introduction

- It is essential to compute motion plans that respect the kinodynamic constraints of a robot.
- Feedback motion planning** considers motion planner and controller in tandem.
- In workspaces with dynamic obstacles, it is necessary to update motion plans on-the-fly.
- Quick replanning** methods derive efficiency by repairing an old plan instead of replanning from scratch.

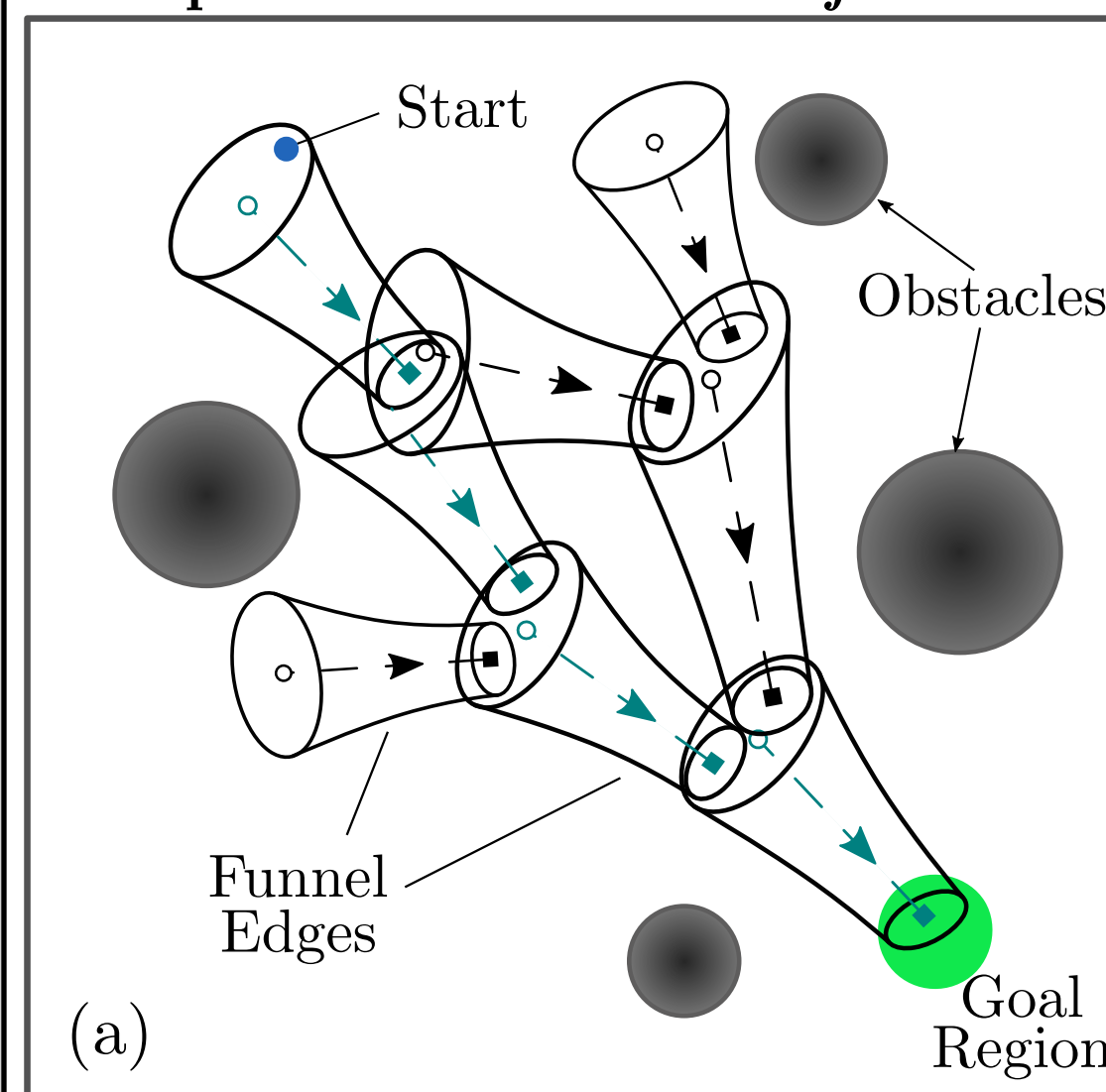
Objective

- We present an **online feedback motion re-planning** algorithm, that is relevant in practical scenarios such as safely navigating through dynamic environments.
- We compute **dynamically feasible** motion plans with formal invariance guarantees for any general nonlinear robot-system.

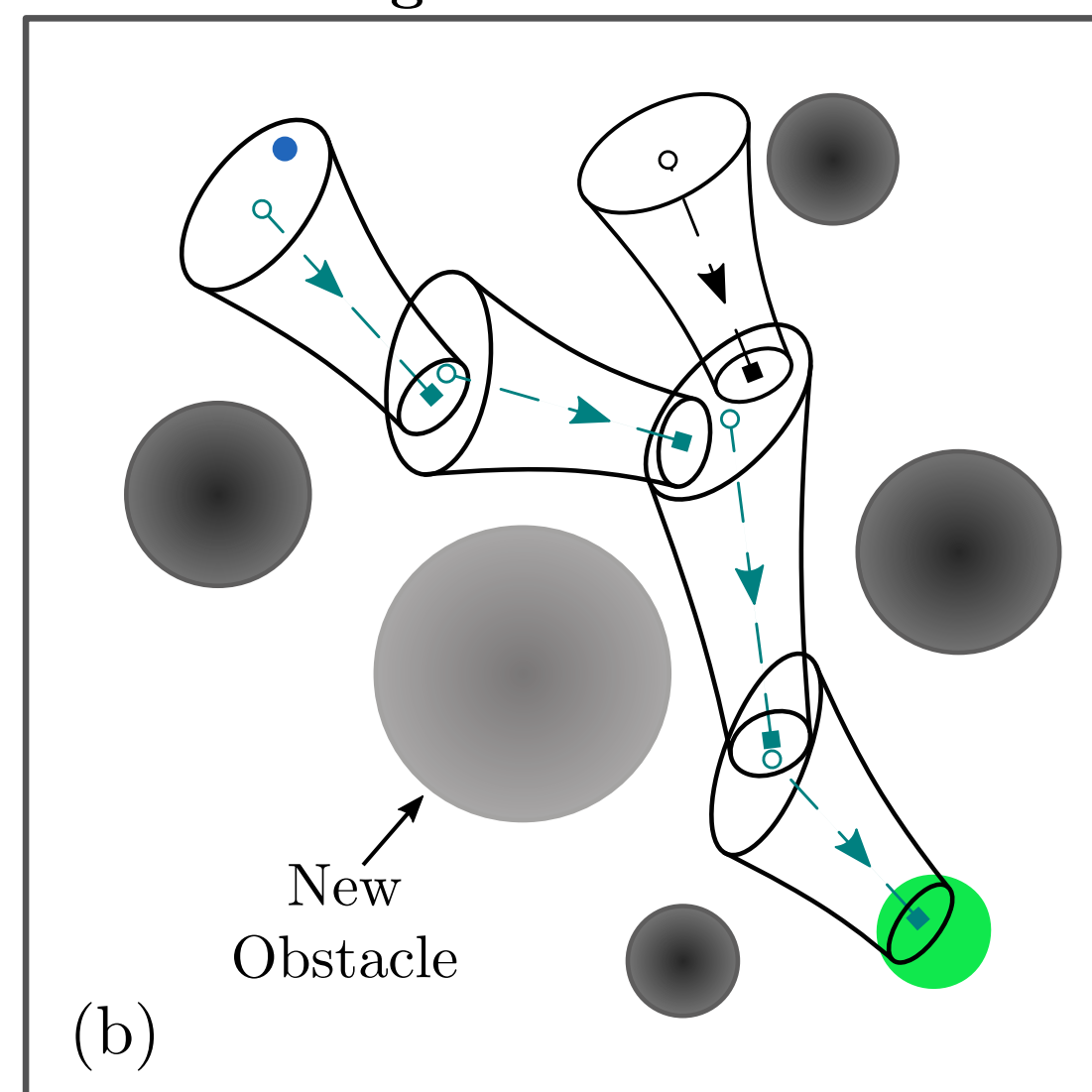
Contributions

- The authors believe this is the first work to propose techniques for **funnel-based online re-planning** using quick graph-rewiring methods.
- We present a **novel graph data-structure** to represent a network of volumetric funnels.

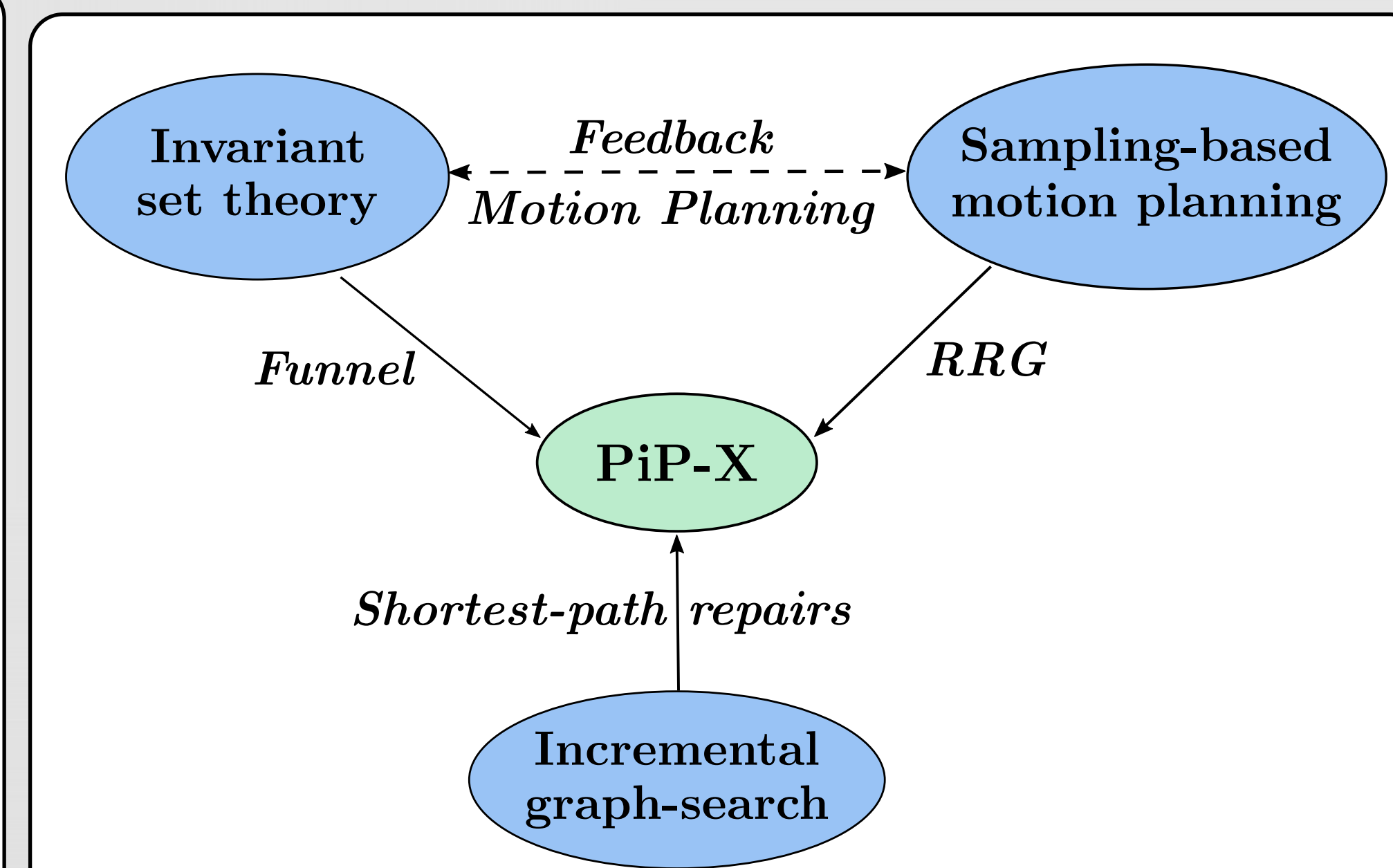
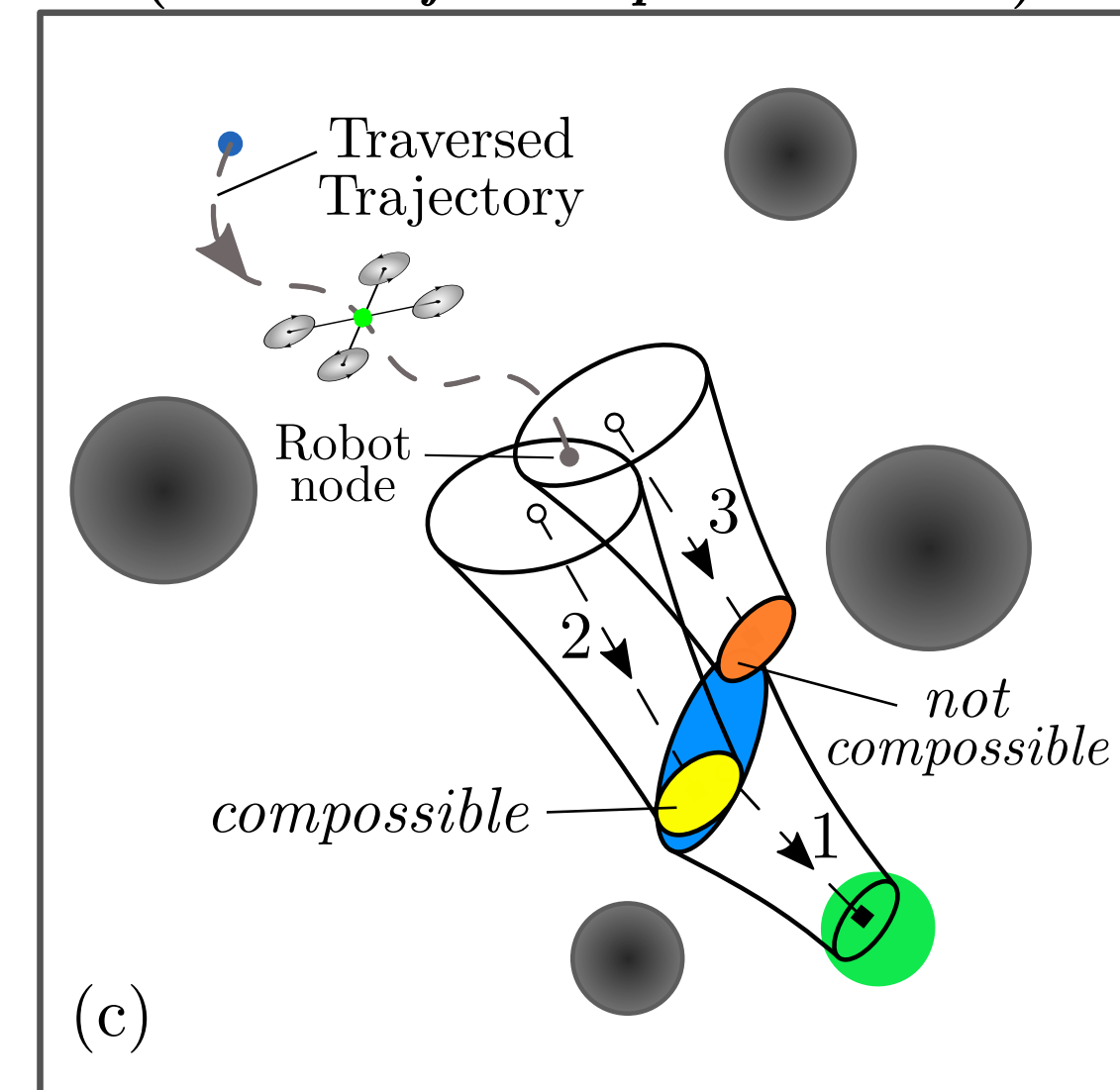
Funnel-graph – a network of sequencible volumetric funnels



Replanned *funnel-path* due to changes in environment



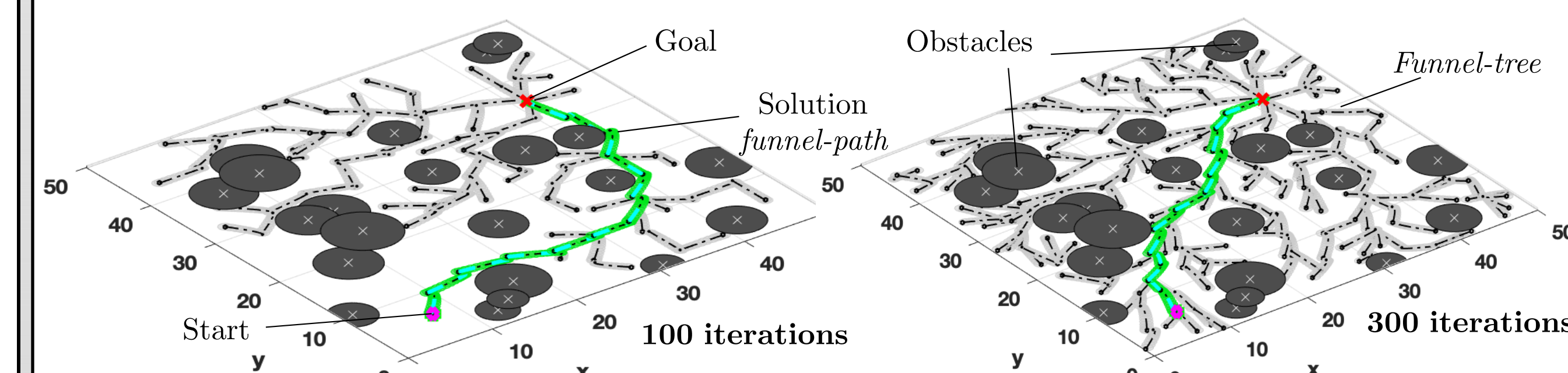
Analysing sequencibility of funnels (solution *funnel-path* is 2 $\rightarrow$ 1)



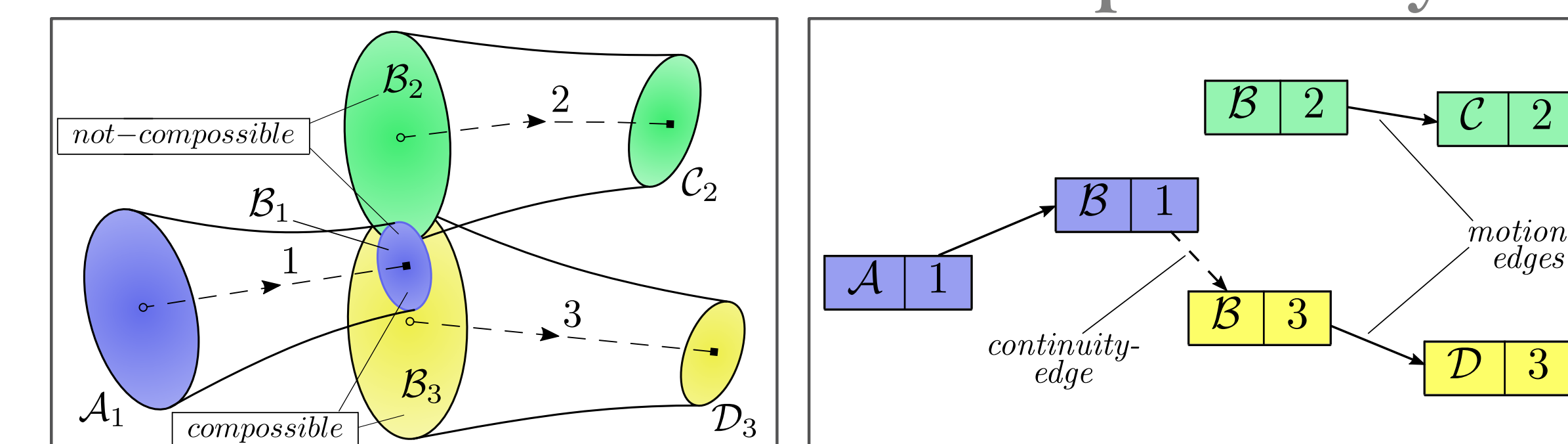
Approach

- Our method consists of an **offline** stage of dynamical systems analysis, and an **online** phase of sampling-based funnel-graph construction and incremental re-planning.
- We pre-construct a library of **backward-reachable invariant sets** using Lyapunov theory, for various combinations of initial and final states.
- A **roadmap of funnels** is iteratively built from the goal using sampling-based techniques and its **connectivity** is represented using an **augmented graph data-structure**.
- Using incremental search and graph-repairs, we keep updating the **shortest funnel-path-to-goal subtree** in the underlying graph.

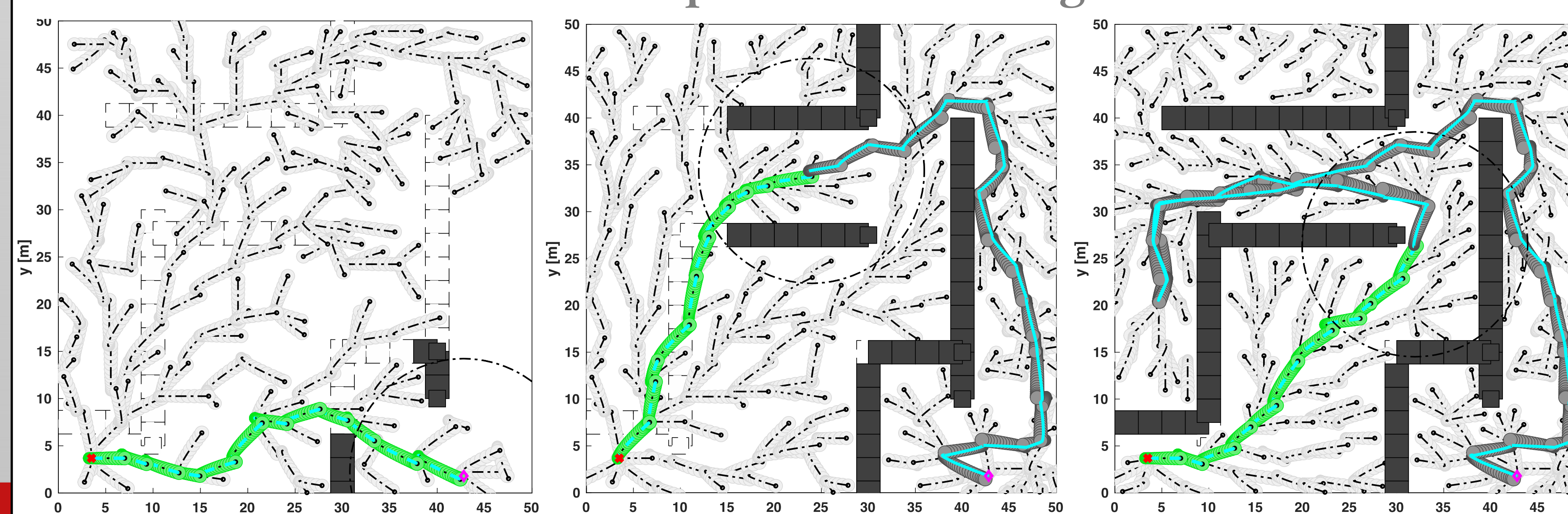
Results in a static random forest environment



Graph data-structure representing the funnel-network and its inter-compossibility



Simulation results of a quadrotor running PiP-X in a maze



References

- Majumdar, A., Tedrake, R.: Funnel libraries for real-time robust feedback motion planning. IJRR 36(8) (2017)
- Koenig, S., Likhachev, M.: D*Lite. AAAI/IAAI 15 (2002)

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